TITLE: Formally Authored Research Proposal and Methodology Framework: *Immersive Extended Reality (XR) Frameworks for Accelerating Advanced Pedagogical Retention in STEM and Computer Science Architectures.*

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DATE OF CREATION: August 22, 2022 (Completed during Graduate Tenure at Minnesota State University Mankato)

Abstract: Computer science is a challenging field that requires students to master many abstract and complex concepts. While standard 2D simulations provide some baseline understanding, many students still struggle due to low retention and a lack of engaging instructional methods. To fill this gap, this study proposes an interactive Virtual Reality (VR) environment designed to provide a hands-on learning experience. Developed using the Unity Engine, this system helps students grasp the practical logic of programming to solve real-world problems and retain that knowledge over time.

Inside this 3D VR environment, students will interact directly with the subject matter using VR headsets. A core feature of this environment is a built-in checkpoint function, which will allow users to save their progress and return to complete challenges to practice them again. This VR system will be tested on a group of 28 computer science students. The anticipated results aim to demonstrate that users in the VR environment will achieve a significantly better understanding of complex programming concepts compared to students learning through traditional, lecture-based methods.

Keywords: Virtual Reality, VR, computer science education, 3D simulation, Human centered computing, interactive learning environment, Unity Engine

Introduction/Background

Virtual Reality (VR) set being economical and available in the last few years. VR as of late has gotten to be a prevalent innovation in numerous settings such as amusement, military, and education [6]. The incorporation of VR in education is one of the popular topics due to its potential and opportunity in the academic field. VR utilizes computer innovation to form a recreated environment which places participants in center of a three-dimensional environment. VR helps give participants to have high level of exploration and involvement and keep them engaged in virtual environment by making them feel the substance of the subject. As of now VR equipment has become affordable, easy to use with simple setup making it easier to have environment setup with affordable budget. VR is being used to simplify the complex problem of the subject in education making it easier for the participants to understand the subject matter. Computer science is one of the most sought-out subjects nowadays, it is just as complex to learn and understand its concepts. Computer science education involves teaching, learning different concepts such as programming, software development life cycle (SDLC), human computer interaction (HCI) and computational thinking. The VR environment provides opportunity for safe practical hands-on learning experience for students [10] to understand complex concepts. Among them this study aims for participants to understand logical thinking behind the complex concept of programming that can be used to solve real world problems and retain the learned knowledge using three-dimensional VR environment. This will help students to increase their understanding of programming and apply learned knowledge in real situations. To achieve this aim, an application to help participants be in 3D environment should be developed using Unity Engine and VR set that supports, help navigate participant inside the environment is needed. The

proposed VR environment will be tested on 28 students in total taking computer science as their major to answer the research questions of the study:

- Will the proposed VR environment make it easier for students to learn and understand the logical thinking behind programming?
- Will the proposed technique help retain knowledge longer?
- Will the proposed VR environment increase the number of students that are able to understand the logical thinking of programming?

The issue raised inside these questions can be tended to by utilizing completely immersive virtual reality environment [8] which is still not possible as of now bringing challenges in this field and the design of the study.

Literature Review

The main concept of VR is immersion and presence of the participants in the simulated environment which Schwan and Buder stated that VR being an effective device that improves the learning procedure by giving spatial, multimodal visualization that's exact, straightforward, and simple to get it [7]. The researcher has found that students perform way better in programming a concept of computer science when visualization is utilized for instructing [9] which is able to provide interactive and engaging environments. During the process of finding other related work, it was found that numerous studies have been done in the topic of education of computer science utilizing virtual reality. There was rise in the studies recently due to achievable VR environment development kit and affordable VR sets available to perform the experiments needed to prove their claim. Some of the literature reviews were done where VR environment was the main focus on

different topics such as programming education, concept of software development life cycle (SDLC), solving real world problem which is discussed in following.

Programming education has become a mainstream subject taught in schools and higher education academy. As numerous studies are attempting to examine within the performance of students in programming courses due to students having difficult time understanding the complex concept of programming such as functions, loops, and recursion. Modern innovations are showing possibilities on educating such concept to students for superior understanding and among them is smart learning environment (SLE). SLE focuses on instructing students by utilizing virtual reality (VR) innovation. The features that make SLE "smart" and separate itself from traditional learning environment is because of its adaptivity, context-awareness, ubiquity, personalization, programmed appraisal, personalized criticism, inspiration, and social-awareness. VR-based architecture aims to help participants learn the concepts of programming through visualization and interaction. The SLE framework prototype obtains requirements and models it by using Unified Modelling Language (UML). It is based on three different layers such as data layer, logic layer and presentation layer respectively. In data layer, the input data from the participants is obtained automatically at every learning session which is then used to support the participants learning progress and personalize their learning experience. In logic layer, the modules contain instructions and program logic in 3D based games to help learn by creating simple games. Finally in presentation layer, smartphone is placed in front of the headset and the participants can control the game object using hand controller or joystick and whereas learning is on progress the learning object naturally adjust to the participant based on set of rules for sequencing learning question and results of the learning evaluation is carried out within the session. UML is utilized to plan to

educate the programming concepts, practice tests at distinctive level with programmed appraisal done to assess the result [1].

The virtual reality (VR) main components are hardware and software which helps the system present virtual environment as the real one to the end user. High-quality visual materials are made by utilizing virtual reality innovation empowers to instruct the progressed hypothetical substance successfully making instruction one of the appropriate areas to apply this innovation. There are more advantages to using VR for education such as increments in student's inspiration, diminished learning curve, improvement of issue tackling abilities, capacity to interact with the disentangled occurrences of complex and costly frameworks securely, interaction of geologically conveyed students and introduction of student-particular educating strategy rather than common reason instructing approach. The system architecture for VR education consists of two different modules, one for instructor application which manages the session and another for student application which connects to the session. Instructor application must be initiated for the learning session to start in which student application will connect to view and experience the session. For the experiment, the students learned using VR environments (2nd group) was compared to students learned using traditional method (1st group). After performed test was performed, the results showed that 2nd group performance was better than 1st group statistically. The students using VR system were found active with their participation in lecture and were interested in learning more about the topic. VR as a subject needs more exploration since not until 2000s were numerous users not able to buy it due to its high cost and only as of late is being accessible for reasonable cost [2] so, there is room for more improvement.

In other review, the framework utilizing virtual reality (VR) was proposed to test on bunch of students to extend their execution in tasks such as coding, testing, necessity gathering and so on

which are part of software development life cycle (SDLC). VR was considered to endure convenience issues but students that learned utilizing VR still gave positive reviews and reveal progressed performance. The framework mechanism has two complementary applications such as desktop application for generating project scenarios and 3D virtual office environment. The desktop application purpose was to enable the authorized people to create different project scenarios that contained the project development process such as requirement gathering, design, implementation, test, and maintenance phases which this application produced in Extensible Markup Language (XML). After the creation of XML file, it was then animated in the VR environment at the beginning of the simulation. The participants then would connect to the VR environment for the experiment to see how much they understand the concept. The results of this experiment were done pre-test and post-test where a semi-structural interview was organized with specialists and students alike to get their supposition on the proposed framework and also summarized in numerical values so that it can be analyzed by comparing the two tests in sequence to produce valid results in terms of statistic. Both qualitative and quantitative data were obtained throughout this experiment so that it would be helpful at the end to analyze and validate the obtained data [3] to find the true result in it.

Virtual reality (VR) is an innovation that gives a productive and valuable preparing environment for individuals. This advancement enriches the members in a virtual environment outlined to be like real ones. The aim is to encounter the issues that individuals may confront in real world in a virtual environment and dodge by making such mistakes. Numerous studies show that VR is a successful preparation instrument that can be utilized to extend individuals' level of information and experience. The agile methodology a concept of software development is hard to be trained for the software development teams since, it is combined of "user centered design" and "model

driven development", difficult to get it without confronting genuine world operation. Indeed, on the off chance that there's individual that caught on the concept, he/she cannot effectively exchange this information to novice individuals who don't completely comprehend the specialized subtle elements of the concept. Literature survey was done to discover the studies that was utilized to train the individual in which studies were found in comparative issues but in numerous subjects such as VR being created for preparing both crane administrators and groundwork force to dodge making botches amid their development work. Such studies show the VR is appropriate for hands-on practice to train individuals in numerous subjects. There was experiment done utilizing Second Life as an instruction apparatus where VR classroom comparative to genuine classroom was made to supply communication between students and lecturer with bunch of students included. The result demonstrates that a VR technique is beneficial in upgrading the information levels of students in the software engineering field. Due to VR environment individuals were able to train and participate without real life risks [4].

Learning computer science may be a challenging assignment and numerous of the concepts and wonders included are troublesome to get it. Programming makes a difference in students' computational thinking abilities, cultivating capacity to think on numerous levels of deliberation. Digital learning applications that aid in seeing the issue have become basic apparatus to assist on comprehending the issue and encourage its learning progress. As virtual reality (VR) innovations offer modern inventive possibilities for interactive and immersive digital learning. VR put itself apart from 2D digital learning because of its features to have individuals in a digitalized world and make those individuals interact with objects directly. The usage of VR in educational means has received positive reviews for being engaging. The experiment was performed implementing A/B split in participating students in a group of 20 by comparing visualization about sorting algorithms

in two different environments such as WebGL for 2D digital learning in website and HTC Vive for VR learning. The web and VR adaptation have same highlight in terms of selecting and controlling sorting algorithms in differentiate to the input strategy as web has keyboard and mouse whereas VR has controller of the HTC Vive. The results were based on engagement, emotions and learning experience, which 17 out of 20 students preferred the VR adaptation over the web adaptation. From the experiment it was found that users find more potential from VR learning due to its interactive and engaging feature. The students found VR involvement more user-friendly, fun, associate, and way better for visualizing the issue on hand [5].

After analyzing the literature review it was noticed that these studies have a few impediments when utilized for educational reasons. The results that were obtained through comparison between students who knows software engineering concepts and who do not did not have a single action as a correct one for each scenario, researchers are not sure of students who did not know software engineering concepts [2] had learned of it through other sources before, after or during the experiment. The research purpose of virtual reality (VR) environment for participants to gain experience to learn about task needs to be learned with challenges and importance of VR for on-hand training [1] but, does not provide plan or explain on how this can be achieved. While the results of the research proved that the student using VR training method performed better than traditional training method, but the algorithms used had limited variations of structured questions [3] that will be helpful for students. The experiment participants had age gap of 17 years between the oldest and the youngest so there is no way of knowing of their learning retention rate [4] making it unsure if the participants will be able to retain what they learned using VR environment.

Methodology

This study will use quantitative approach since it will obtain numerical data through numerous tests. The methodologies are derived from [3] to achieve the research aim of this study. This study will have three different tests T1, T2 and T3 for the participants to observe their progress of knowledge. T1 test will take place before participants engage in learning T2 will take after participants are done with the learning using VR or traditional environment and T3 will take place after a week of the completed learning. T1 will help to gauge the current level of understanding of the subject of the students before attending the experiment while T2 will help gauge their level of understanding of the subject after finishing the engaged learning and T3 will help gauge their level of understanding of subject after interval of time from their experience with the VR or traditional environment. The value that will be obtained from these three different tests will be analyzed using statistical methods in order to figure out the difference between and compare the results of T1, T2, T3. The Unity Engine is selected for developing the application for engaging VR environment and oculus quest set will be needed to navigate within the created VR environment. The experimental design will be set up which will have 14 students each divided randomly to use 3D VR environment and 2D environment having computer science as their major to learn the concept of computer science. There are certain criteria of selecting participants:

- The participants' age should be between 18 - 25 years.
- The participants should be at same education level.

The experiment will last for 5 weeks, such as 4 weeks for engaged learning and last week being interval time for T3 to take place at the end of the day of week 5 to learn about how the participant are able to retain their knowledge. While traditional environment users use the 2D

simulation whereas VR environment will use the three-dimensional simulation. The VR environment will have a checkpoint function that will let participants save their progress until they logged in and save the problem, they want to go to directly later so they can learn again on how to solve it.

Evaluation of Results

The three research questions will provide an important analytical framework for the data collected. In the beginning, before the learning process a test (T1) will be taken to gauge the understanding of the concept of the participating students by solving the number of questions regarding programming and concept of programming. The data will be obtained from 28 computer science students that will be divided randomly on two groups with 14 students each using VR environment and traditional environment which will be plotted in bar chart with each group respectively. Similarly, after the learning process the test (T2) will be taken to gauge the progress understanding of the concept of the participating students by solving question as T1 test while the question will be different than T1 test, it will have same pattern for the question that will need to be solved and are able to obtain data. While the bar chart will be updated with T2 data in already plotted T1 data bar chart. Finally, after a week of interval from the completion of the learning process the test (T3) will be taken to gauge how much the participating students are able to retain their knowledge by solving the question similar in pattern from T1 and T2 from which data is obtained and bar chart will be updated in already populated T1, T2 data with T3 data respectively.

The data obtained through three different test T1, T2, T3 will have differences and comparison between these data will be made to answer the research question of this study such as if there is increase in number of students understanding the logical thinking behind programming using VR

environment than traditional environment, is learned knowledge is retained longer when using VR environment and is it easier for learn understand logical thinking behind programming using VR environment. Participants having learned the concept of programming before experiment will not affect the data obtained from the test since, all the participants will be divided randomly and have background in computer science with same education level which is expected of them to be exposed to certain level of concept of logical thinking behind the programming.

Ethical Considerations

Given the nature of this study, the participation of people is needed so there are some ethical issues and rules that need to be taken into consideration. First the participants will need to be treated with respect and courtesy. A technique of informed consent will be adopted for confidentiality regarding experiment details and data collection of all the participants. The participation will be voluntary as they need to invest a lot of time since the experiment will last 5 weeks. The participants will be within the age of 18 – 25 years. The participants will be let know that they can withdraw from participation in experiments whenever they want to without having to face consequences for their withdrawal. The students will be given full explanation regarding the research time period and summary of the research experiment on what their role is going to be in it so that they can decide voluntarily without pressure. The consent form that participant will need to sign will follow the guideline of Minnesota State University Mankato IRB for participant consent information accordingly. The participants will need to fill the form for their involvement in the research, mentioning that they were informed and understand the detail of the research experiment, which won't be an issue with their participation later. The informed consent form will also include the confidentiality and anonymity of the participants data collected during

research and will be used in the research only, no identity will be mentioned during reporting of the results.

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Appendix I

Anticipated Funding Requirements

Oculus Quest 2 set - $299.00 (each)

This set is built for virtual reality and will be needed to be connected in virtual reality (VR), navigate, explore inside the 3D VR environment.

Unity Engine - $1800/year per seat for teams and individuals (Free for student enrolled in accredited educational institution)

This software is needed to create a 3D VR environment that will be used for research experiments.

Anticipated IRB and Other approvals

Application for the Conduct of Human Research Form provided in IRBNet by Minnesota State University Mankato IRB library.

Thesis/APP Committee

Dr. Alkhushayni Suboh, Associate Professor – Chair of the thesis/App committee

Ph.D. in Engineering – Computer Science, University of Wisconsin-Milwaukee

M.Sc. of Science – Computer Science, University of Wisconsin-Milwaukee

Professor Dr. Alkhushayni Suboh has knowledge of developing attractive user interface which can be embedded for the 3D environment for engaging, interactive learning. Dr. Alkhushayni has enough research experience and knowledge regarding virtual reality to guide me through and oversee this study.

Some of the research interests of Dr. Alkhushayni are:

- Action research and Virtual Reality.
- Networking and Security.
- Human Computer Interaction (HCI)
- Artificial Intelligence
- Healthcare Solutions

Dr. John Burke, Assistant Professor – member of the thesis/APP committee

Ph.D., University of Illinois at Urbana-Champaign

Professor Dr. John Burke has knowledge of developing environments using Unity Engine which is crucial to the research method for creating a 3D environment. Dr. John Burke has enough experience, knowledge, and skills on topic of virtual reality which will be of help to me to ask for guidance on this study.

Appendix II

Tentative Timetable for Completion of the Thesis/APP

Include dates for:

- completion of research
- first draft of thesis (at least six weeks prior to the defense)
- final draft (at least two weeks prior to the defense)
- defense (confirm the date with the chair of the thesis/APP committee)

Research Project/Thesis/APP Stage	**Completion Target Date**
Literature review	07/06/22
Finalize research questions	09/10/22
Finalize the thesis/APP committee	10/25/22
Construct research design	12/20/22
Conduct experiment/gather data	03/10/23
Analyze data	05/15/23
Draft thesis to advisor(s)	06/05/23
Schedule defense	08/02/23
Final thesis submission for signatures:	10/09/23

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